To provide a production-grade DevOps architecture, I have designed a template based on a **modern containerized backend (e.g., Python/FastAPI, Node.js, or Go)**.

Since you did not specify a language, I am assuming a standard **Node.js/TypeScript** or **Python** backend.

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1. Deployment Architecture

- **Pattern:** Micro-services ready, container-based.
- **Infrastructure:** AWS (or GCP/Azure equivalent).
- **Components:**

* **VPC:** Private subnets for application/DB, public for Load Balancer.

* **Load Balancer:** AWS ALB (Application Load Balancer) to distribute traffic.

* **Compute:** AWS ECS (Elastic Container Service) on Fargate (Serverless containers).

2. Docker (Containerization)

Use a **Multi-stage build** to keep images small and secure.

```
# Stage 1: Build
FROM node:20-alpine AS builder
WORKDIR /app
COPY package*.json ./
RUN npm install
COPY . .
RUN npm run build

# Stage 2: Production
FROM node:20-alpine
WORKDIR /app
COPY --from=builder /app/dist ./dist
COPY --from=builder /app/node_modules ./node_modules
EXPOSE 3000
CMD ["node", "dist/main.js"]
```

3. CI/CD (GitHub Actions)

Automated pipeline triggered on `main` push:

1. **Test:** Run unit/integration tests.

2. **Build/Push:** Build Docker image and push to AWS ECR (Elastic Container Registry).

3. **Deploy:** Trigger AWS ECS service update.

4. Hosting & Database

- **Compute:** **AWS ECS Fargate** (No server management required).
- **Database:** **AWS RDS (PostgreSQL/MySQL)** configured in a Multi-AZ deployment for high availability.
- **Caching:** **AWS ElastiCache (Redis)** for session management and performance.

5. Environment Variables

Never store secrets in source control. Use **AWS Secrets Manager** or **Parameter Store**.

- `DB_URL`: Endpoint for RDS.
- `REDIS_URL`: Endpoint for ElastiCache.
- `JWT_SECRET`: Loaded at runtime via Secrets Manager.
- `NODE_ENV`: 'production'.

6. Monitoring & Logging

- **Logging:** **CloudWatch Logs**. Centralized logs with retention policies (e.g., 30 days).
- **Monitoring:** **AWS CloudWatch Metrics + Prometheus/Grafana**. Monitor CPU, Memory, and request latency (p99).
- **Alerting:** CloudWatch Alarms for 5xx errors or CPU utilization > 80%.

7. Scaling

- **Horizontal Pod Autoscaling:** Configured in ECS.

* **Trigger:** CPU utilization > 70%.

* **Constraints:** Minimum 2 tasks (for redundancy), Max 10 tasks.

- **Database:** Use Read Replicas if read-heavy traffic increases.

8. Backup Strategy

- **RDS:** Automated daily snapshots (stored for 7 days) + Point-in-Time Recovery (PITR) enabled.
- **Infrastructure:** All infrastructure is managed via **Terraform** (Infrastructure as Code). If the region goes down, the entire stack can be redeployed to a new region within minutes.

Summary Checklist for you:

1. [ ] **Terraform/Pulumi:** Start writing your infrastructure code.

2. [ ] **Secrets Manager:** Move all your `.env` variables there.

3. [ ] **Networking:** Ensure your Database is in a **Private Subnet** and not accessible from the public internet.

4. [ ] **Security:** Implement an IAM role for the ECS task with minimal permissions (Principle of Least Privilege).

Would you like me to generate a specific Terraform script for this stack or refine it for a specific programming language?